

AI AND ML OPTIMIZATION

The game-changer reshaping financial services



Face the Future with Confidence

For financial services firms:



By 2026 **>80%** enterprises would have invested in GenAI APIs and Models



As per a government survey, more than **80%** of financial institutions have implemented chatbots to improve customer service.



The global AI in fintech market is projected to reach **\$22.60 bln** by 2026.



Furthermore, **65%** have used fraud detection AI engines, making it the subsequent most frequent use case in the financial services business

With intensifying competition, now's the time to integrate AI and ML for exponential surge and competitive advantage.

AI ML Meaning

At Protiviti, we see artificial intelligence (AI) not merely as automation, but as the driving force behind groundbreaking creativity, metamorphosis, and an unprecedented surge forward. We believe that harnessing the limitless potential of artificial intelligence is essential today to revolutionize your business and stay ahead of the competition.

But what is AI & ML?

Put simply, AI and ML refers to set of technologies used to simulate human intelligence in machines/systems. AI & ML by itself is not a system, but it's used in systems to perform tasks that may typically require human-like abilities such as learning from mistakes, understanding, and responding to spoken/written language, problem-solving etc.

AI ML in FSI

In today's fast-paced and data-driven world, the financial services industry is undergoing a profound transformation, driven in large part by the adoption of artificial intelligence (AI). AI is essential in revolutionizing the financial landscape and the modern world is adapting to its capabilities through:



Prediction and Forecasting



Customer Insights and Personalization



Data Analysis and Pattern Recognition



Adaptive learning Analysis and Pattern Recognition



Automation and Optimization



Natural Language Processing (NLP)



Fraud Detection and Security

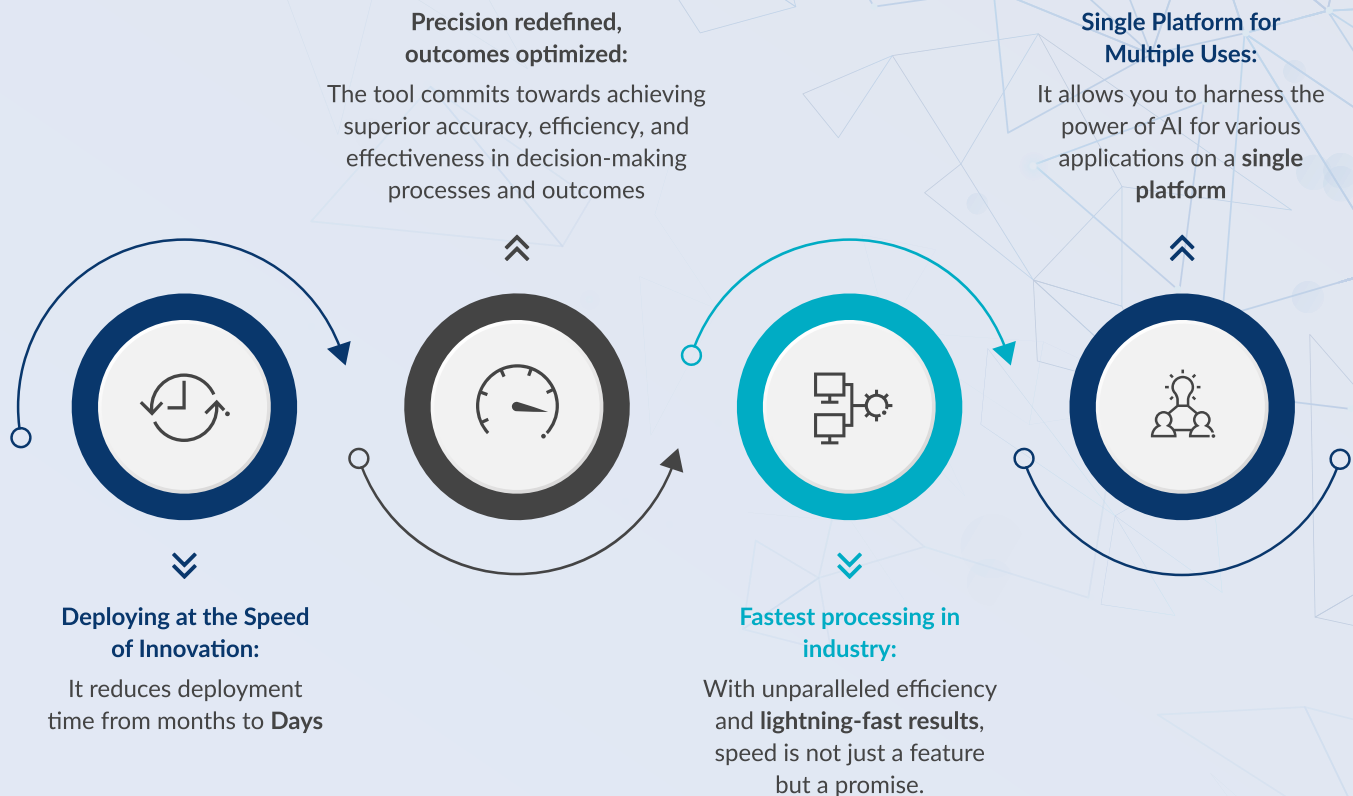
Highlighting the vast capabilities of AI and ML, yet realizing their full potential demands an optimized platform to harness their power effectively.

Protiviti Tool

Introducing our revolutionary AI and ML platform - a versatile solution transcending traditional boundaries.

Protiviti's AI/ML tool tailored for financial services can serve as the cornerstone for innovation, transformation, and unparalleled growth, driving the industry towards a future defined by efficiency, adaptability, and strategic foresight. Protiviti's AI/ML tool is powered by cutting-edge Artificial Intelligence and Machine learning algorithms. From data challenges to complex business dilemmas, unleash its transformative power for comprehensive problem-solving.

Protiviti Tool Advantages



Use cases and Potential Applications:



Predictive Analytics: Potential applications

- **Credit Risk assessment:** Predicting Probability of Default (PD) for Credit Risk Assessment, which assesses the possibility of borrowers defaulting on specific loan types.
- **Market Volatility Prediction:** Forecast volatility, risk and returns in specific financial instruments or markets to optimize bank's investment strategy.
- **Operational Failure Prediction:** Predicting failures or disruptions in financial operations, such as ATM networks or internet platforms.
- **Customer Lifetime Value (CLV) Prediction:** Determining the future worth of specific customer groups based on their lifetime association with the bank.
- **Transaction Fraud Detection:** Detecting certain fraudulent activity in banking transactions, such as identity theft or card skimming.



LLM models: Potential Applications

- **Regulatory Compliance:** Interpreting and analysing regulatory documents and guidelines from sources like Central bank websites to ensure compliance with financial regulations and reporting requirements
- **Sentiment Analysis:** Evaluating customer sentiment from social media, customer reviews, and other textual data sources to understand public perception and identify potential opportunities
- **Market Intelligence:** Extracting insights from news articles, research reports, and other textual sources to form investment strategies and market assessments
- **Customer Support Automation:** Automating customer support processes through chatbots and virtual assistants powered by LLMs, providing quick and accurate responses to customer enquiries
- **Analytical Insights:** Gen AI platform to give analytical insights to business owners



Anomaly Detection: Potential applications

- **Fraud Detection:** Detecting fraudulent transactions, account takeovers, and other suspicious activity
- **Anti-Money Laundering (AML):** Identifying any anomalous trends in financial transactions that could suggest money laundering or terrorist financing operations
- **Network Security Monitoring:** Identifying any odd network traffic or access patterns that could suggest unwanted access or cyber threats
- **Credit Card Fraud Detection:** Detecting illegal transactions or fraudulent behaviours related to credit card use, such as card skimming or identity theft
- **Monitoring ATM transactions** for unusual withdrawal trends, card cloning, or manipulation



Clustering and Classification: Potential applications

- **Customer segmentation:** Segmenting the market using demographic, behavioural, or transactional attributes to adapt marketing strategies and personalize offerings
- **Credit Scoring:** Classifying loan applicants into risk categories using past data and credit scoring models to predict their creditworthiness
- **Fraud Detection:** Using anomaly detection and classification techniques to categorize transactions as fraudulent or non-fraudulent
- **Churn Prediction:** Segmenting consumers into churn and non-churn categories based on usage patterns, satisfaction scores, and engagement levels
- **Product Recommendation:** Offering banking products or services to consumers based on their preferences, previous interactions, and similarities to other customers in the same cluster

Though these examples offer a glimpse, the boundless potential of applications knows no limits.

Cases in Point



Client

A micro-lending company in India

An education app startup

Agro firm backed by National Science Foundation



Problem

A backlog of 18,000 loan applications per month

Witnessed a significant growth in customers

Difficulty in predicting yields a farmer is likely to get from its fertilizer



Services

Classification models, Data Merging, Data Enrichment, Statistical Work

Classification models, Data Transformation, Data Visualization, Zapier Automation

Regression Model, Data Standardization, Class Balancing, Outlier Management



Benefit

The tool helped **reduce process time** for each application from **20 minutes** to **5 minutes**

Increase in **retention rate** by **5X**

- Increased crop yield by 5-10 bu/acre
- Increased sales by 11.2%



About US

Protiviti (www.protiviti.com) is a global consulting firm that delivers deep expertise, objective insights, a tailored approach and unparalleled collaboration to help leaders confidently face the future. Protiviti and its independent and locally owned Member Firms provide clients with consulting and managed solutions in finance, technology, operations, data, analytics, digital, governance, risk and internal audit through its network of more than 85 offices in over 25 countries.

Named to the 2023 Fortune 100 Best Companies to Work For® list, Protiviti has served more than 80 percent of Fortune 100 and nearly 80 percent of Fortune 500 companies. The firm also works with smaller, growing companies, including those looking to go public, as well as with government agencies. Protiviti is a wholly owned subsidiary of Robert Half (NYSE: RHI). Founded in 1948, Robert Half is a member of the S&P 500 index.

A woman with long dark hair, wearing a grey blazer over a light-colored top, is shown in profile, looking upwards and to the left. She is wearing a black VR headset. Her right hand is raised, with her index finger pointing towards a glowing digital interface. The interface is composed of various elements: a large, semi-transparent globe in the upper left; a network of blue lines and dots resembling a constellation or data map; and several floating numerical values such as 0.41, 1.66, 10.49, 4.58, 5.56, 0.87, 4.38, 2.12, 0.27, 7.74, 10.18, 10.14, 5.56, 7.59, 8.02, 10.06, 6.09, 0.04, 10.33, 0.237, 5, and 1.88. The text "Face the Future with Confidence®" is prominently displayed in a white, serif font in the upper left quadrant. The overall color scheme is dark blue and black, with glowing blue and white highlights from the digital elements.